

## Bubble Sort

Bubble Sort is a simple sorting algo that repeatedly compares two adjacent elements and swap them if they are in the wrong order.

After every pass, the largest element moves to its correct position at the end of the array.

### Working principle

- Compare two adjacent element
- if the left element is greater than the right element, swap them.
- Move to the next pair.
- Continue until the end of array
- Repeat the same process for the remaining unsorted part.
- Stop when the array becomes sorted.

### Algorithm

Step-1 Start from the <sup>first</sup> index

For (int  $i=0$ ;  $i < n-1$ ;  $i++$ )

Step-2 Compare adjacent element

$arr[i] > arr[i+1]$

Step-3 If they are in the wrong pair swap them

$swap(arr[i], arr[i+1]);$



Step-4 Continue comparing untill the sorted part ends.

for (int j=0; j<n-i-1; j++)

Dry Run: — Array = 13 46 24 52 20 9

Pass 1      13    46    24    52    20   9

13    46    → No swap

46    24    → swap

13    24    46    52    20   9

46    52    → No swap

52    20    → swap

13    24    46    20    52   9

52    9    → swap

13    24    46    20   9   ~~52~~

Pass 2      13    24    46    20   9   52

13    24    → No swap

24    46    → No swap

46    20    → swap

13    24    20    46    9   52

46    9    → swap

13    24    20   9   46   52



Pass 3      13      24      20      9      46      52  
             24      20      → swap  
             13      20      24      9      46      52  
             24      9      → swap  
             13      20      9      24      46      52

Pass 4      13      20      9      24      46      52  
             20      9      → swap  
             13      9      20      24      46      52

Pass 5      13      9      20      24      46      52  
             13      9      → swap  
             9      13      20      24      46      52

Program

```
void bubbleSort (vector<int> &arr, int n)
{
    for (int i = 1; i < n; i++)
    {
        for (int j = 0; j < n - i; j++)
        {
            if (arr[j] > arr[j + 1])
            {
                swap (arr[j], arr[j + 1]);
            }
        }
    }
}
```